

A Response-Blind Fresh Holdout for a Finite Prime-Shell Tilt

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Abstract

TPC-364 identified $\text{beta}=2$ as the only zero-failure member of a finite prime-shell tilt menu, but that phase point was measured on a reused panel. We freeze $\text{beta}=2$ and test it on a response-blind fresh panel. Fifty-one candidate origins are scored using only unsigned weighted square geometry on 256-point pilots; a declared greedy separation rule selects (413342, 410258, 416940). Only then do we evaluate signed matrices. The complete replay contains 384 rows: two betas, three origins, two counts, four shell anchors, two kernel exponents, and four fixed sign laws. The inherited finite spectral cap 0.64 is exceeded in 30 of 192 $\text{beta}=0$ rows and in zero of 192 $\text{beta}=2$ rows. The $\text{beta}=2$ maximum normalized spectrum is 0.61633188509480319, differing from the TPC-364 value by only $4.4345466941875245 \times 10^{-5}$. This is finite, geometry-selected transfer evidence. It is not an asymptotic operator estimate, a source-valid arithmetic normalization, a Route-A/Route-B pass, or a twin-prime result.

1 Question and scope

The preceding TPC-362 shell ladder found that the inherited normalized cap 0.64 first failed at larger shell scales. TPC-363 showed that the failure survived two targeted five-percent restrictions. TPC-364 then introduced the explicit prime weight

$$w_{p,\beta} = \left(\frac{p}{Q}\right)^\beta \quad (1)$$

and found a finite $\text{beta}=2$ phase point on a reused panel. The present paper asks the smallest follow-up question: does the predeclared $\text{beta}=2$ rule remain below the same working cap on new origins?

The word “holdout” is qualified here. The signed response and sign laws are held out from the selection, but the origins are deliberately selected by an unsigned geometry functional. Thus this is response-blind finite transfer evidence, not a probabilistic independent-sample or uniform-in- origin theorem. No source vector, adaptive sign, or arithmetic reassembly is used. The official Session-named Route-A and Route-B evaluator files are absent from this checkout; the local Bridge-B checker is fail-closed evidence only.

2 Finite operator and selection rule

For $I = [x, x + N - 1] \cap \mathbb{Z}$, let $Q < p \leq 2Q$ range over the literal prime shell and define

$$B_p(u, t) = p \frac{66^{2s}}{(66^2 + (u - t)^2)^s} \left(\mathbf{1}_{p|u-t} - \frac{1}{p-1} \right) \mathbf{1}_{u \neq t} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid t}. \quad (2)$$

For a fixed sign law ε , set

$$A_{\beta,\varepsilon} = \sum_{Q < p \leq 2Q} \varepsilon_p w_{p,\beta} B_p, \quad G_{\beta,u} = \sum_{Q < p \leq 2Q} \sum_{t \in I} (w_{p,\beta} B_p(u, t))^2, \quad (3)$$

and normalize by the symmetric congruence

$$A_{\beta,\varepsilon}^\# = D_{G_\beta}^{-1/2} A_{\beta,\varepsilon} D_{G_\beta}^{-1/2}, \quad D_{G_\beta} = \text{diag}(G_{\beta,u}). \quad (4)$$

Table 1: Beta comparison on the fresh finite panel (192 rows per beta).

β	max spectrum	cap failures	max Schur	min effective fraction
0	1.6398827540	30	1.7173737389	1.0000000000
2	0.6163318851	0	0.6454484064	0.6693830009

Table 2: Maximum normalized spectrum by shell anchor.

Q	beta=0		beta=2	
	max spectrum	failures	max spectrum	failures
80	0.6265534342	0	0.3253837005	0
128	1.3238303046	6	0.5444783639	0
256	1.5433463891	12	0.5820069388	0
512	1.6398827540	12	0.6163318851	0

The geometry is a sum of rational squares in the exact finite model. Once it is positive, the two elementary finite envelopes are

$$\|\cdot\|_2 T \leq \max_u \sum_t |T(u, t)|, \quad \|\cdot\|_2 T \leq \left(\sum_{u,t} |T(u, t)|^2 \right)^{1/2}. \quad (5)$$

The response-blind selection uses candidate origins $a_j = 410001 + 257j$ for $0 \leq j < 51$. On the pilot interval $[a_j, a_j + 255]$, calculate beta=2 geometry for each $(Q, s) \in \{80, 128, 256, 512\} \times \{1, 2\}$ and define

$$S(a_j) = \max_{Q,s} \frac{\max_u G_{2,u}}{\min_u G_{2,u}}. \quad (6)$$

Candidates are sorted by decreasing score, with increasing origin as the tie-break, and greedily retained when at least 2048 away from every retained origin. The resulting ordered triple is

$$(413342, 410258, 416940). \quad (7)$$

The rule reads no signed matrix or source response. After the triple is frozen, beta=0 is added as a control and all four laws are evaluated.

3 Finite audit

The protocol uses counts 256, 512, shell anchors 80, 128, 256, 512, exponents 1, 2, and the four fixed laws all-plus, alternating shell index, the prime modulo-four character, and a half-shell split. There are $2 \cdot 3 \cdot 2 \cdot 4 \cdot 2 \cdot 4 = 384$ rows, with a true spectrum recorded for every law. The value 0.64 is an inherited finite working threshold, not a claim uniform in Q .

Table 2 shows the shell-scale comparison. Each entry aggregates three origins, two counts, two exponents, and four laws.

The beta=2 maximum is below the working cap and its maximum Schur value is below the finite working Schur threshold 0.83. Its minimum effective shell fraction is 0.66938300094026681, a diagnostic showing that the finite observation is not being described as a one-prime truncation. Relative to the TPC-364 beta=2 maximum 0.61628753962786131, the fresh-panel difference is 0.000044345466941875245, below the predeclared tolerance 0.001.

The beta=0 failures remain visible on the same origins and settings. This paired control is useful evidence that the transfer is not merely a change in the row census, but it does not identify the source-valid normalization.

4 Exact and independent verification

At $Q = 4$, exponent one, the half-open exact interval $[413372, 413385)$ has shell $\{5, 7\}$. For beta=0 and beta=2, the package stores exact rational matrix and geometry digests. Symmetry is checked by exact equality and every geometry entry is positive.

The producer accumulates prime blocks in increasing shell order. An independently written checker rebuilds the prime sieve, the response-blind selection, the masks, weights, four laws, geometry, Schur/Frobenius envelopes, and

all 384 true spectra while traversing each shell in reverse order. It compares row keys, weights, geometry extrema, matrix metrics, phase counts, and exact anchors. A separate adversarial checker rejects 19 mutations of protocol fields, selection, row counts, phase counts, and claim firewall values while preserving the baseline digest. Normal and optimized executions are required to have empty stderr and byte-identical stdout by the local Bridge-B checker.

5 Claim firewall and route decision

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TPC365_GEOMETRY_SELECTION = PROVED_EXACT_FINITE_RESPONSE_BLIND
TPC365_WEIGHTED_GEOMETRY_POSITIVITY = PROVED_EXACT_FINITE
TPC365_FINITE_REPLAY = NUMERICALLY_CERTIFIED_FINITE_384_ROWS
TPC365_BETA2_HOLDOUT = NUMERICALLY_CERTIFIED_FINITE_SCOPED
TPC365_BETA2_CAP_TRANSFER = NUMERICALLY_CERTIFIED_FINITE_SCOPED
TPC365_BETA2_ASYMPTOTIC_REPAIR = OPEN
TPC365_NORMALIZATION_SOURCE_VALIDITY = MODELING_CHOICE_OPEN
TPC365_GROWING_OPERATOR_BOUND = OPEN
TPC365_SOURCE_UNIFORM_L2 = OPEN
TPC365_ARITHMETIC_ADVANCE = NO
TPC365_FIXED_POWER_CREDIT = 0
TPC365_FULL_GATE_B = OPEN
TPC365_TWIN_PRIME_RESULT = NONE

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The strongest positive result is a fixed-rule, response-blind finite transfer on three new origins. The strongest obstruction is that the origins are still geometry-selected and the result has no growing-scale or source arithmetic control. Therefore no Route-A or Route-B gate is closed, no fixed power is credited, and no twin-prime conclusion follows. The next minimal test is to keep beta=2 fixed and extend the shell ladder beyond $Q = 512$ on a new declared scale panel. A failure should be retained as a scale obstruction rather than repaired by refitting beta.

6 Conclusion

TPC-365 removes one immediate objection to the TPC-364 phase point: the predeclared beta=2 rule survives a fresh response-blind finite panel, while the beta=0 control retains the inherited high-shell failures. The result maps a more promising branch of the finite route, but it does not bridge the arithmetic source problem. Growing- Q control, source-uniform L^2 , reassembly, and the twin-prime endpoint remain open.

TPC365_ARITHMETIC_ADVANCE=NO, TPC365_FIXED_POWER_CREDIT=0, TPC365_FULL_GATE_B=OPEN.